

	Document Type:	PLANT MODEL DOCUMENTATION	Document Number
	Description (English)	TRANSIENT ELECTROMECHANICAL BPM MOTOR	W12345678
	Description (Local)		Revision B
Whirlpool Corporation, Benton Harbor, MI, USA			

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1. PURPOSE

This document describes the transient electro-mechanical model for Brushless Permanent Magnet Motor (BPM) and how to apply them in various simulation environments.

2. SCOPE

This library includes global models of BPM motors. The application can be used in motor control algorithm development for various platforms (laundry, dish, refrigeration, dryer) with the usage of MIL, SIL, PIL, RT-PIL or HIL.

3. TERMS, DEFINITIONS, ABBREVIATIONS, AND ACRONYMS

TERM	DEFINITION
BPM	Brushless Permanent Magnet Motor
MIL	Model in the Loop
SIL	Software in the Loop
PIL	Processor in the Loop
RT-PIL	Real-Time Processor in the Loop
HIL	Hardware in the Loop

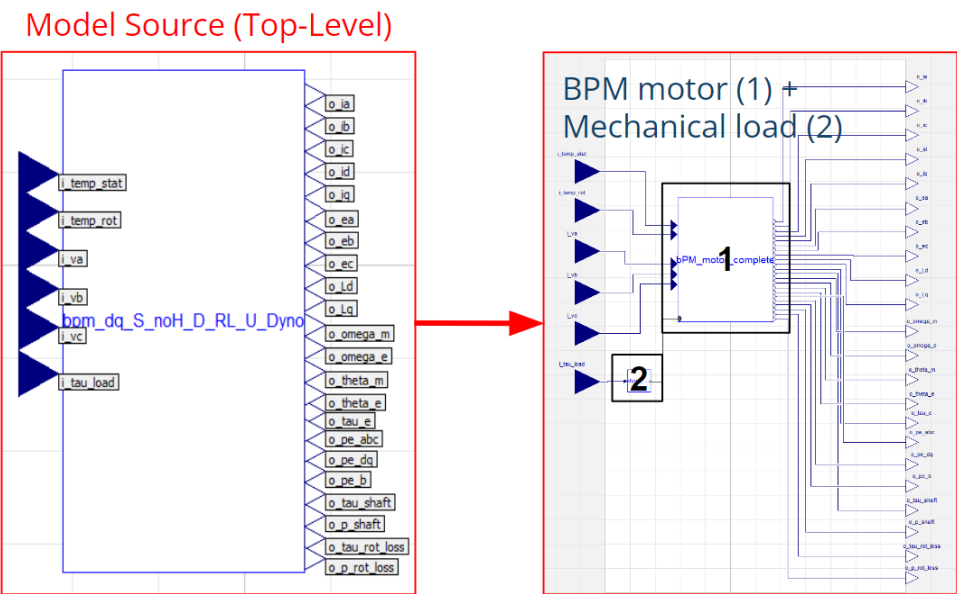
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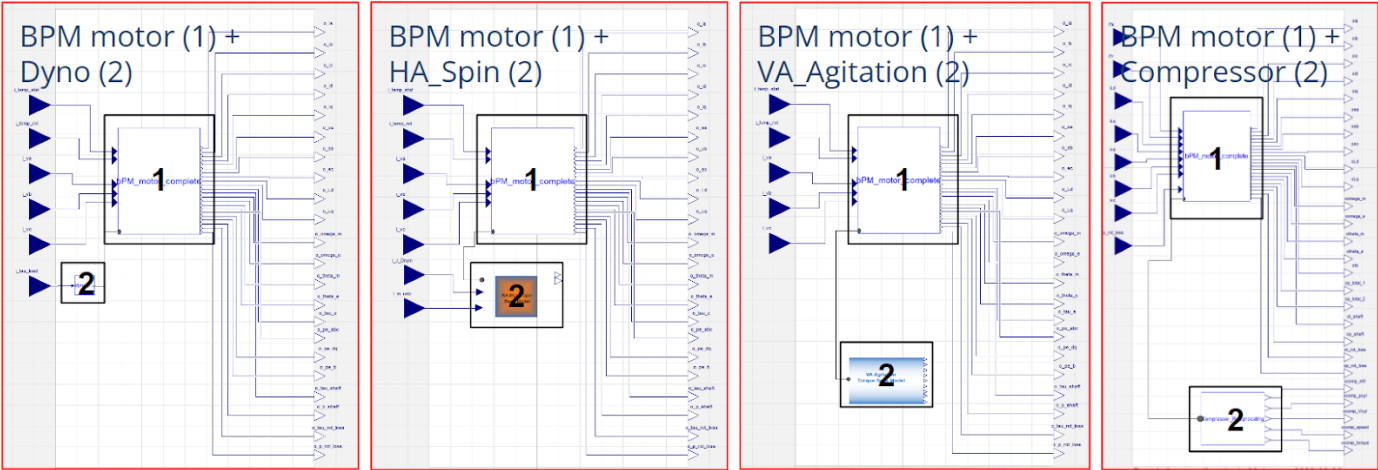
4. PLANT MODEL IN MODELICA

4.1 ARCHITECTURE

At the top level, all model variants follow the same architecture pattern. The model consists of a complete BPM motor electro-mechanical transient model (1) and a mechanical load model (2).



Top-Level variants due to mechanical load



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4.2 INTERFACE

The core interface remains the same for all top-level model variants.

Inputs			
Signal name	Description	Unit	Data type
i_va	Input voltage, phase a	V	Real
i_vb	Input voltage, phase b	V	Real
i_vc	Input voltage, phase c	V	Real
i_tau_load	Load torque	Nm	Real
i_temp_stat	Stator temperature	degC	Real
i_temp_rot	Rotor temperature	degC	Real

Outputs			
Signal name	Description	Unit	Data type
o_ia	Output current, phase a	A	Real
o_ib	Output current, phase b	A	Real
o_ic	Output current, phase c	A	Real
o_id	Output current, axis d	A	Real
o_iq	Output current, axis q	A	Real
o_Ld	Stator inductance, axis d (debug output)	H	Real
o_Lq	Stator inductance, axis q (debug output)	H	Real
o_ea	Back-electromotive force, phase a	V	Real
o_eb	Back-electromotive force, phase b	V	Real
o_ec	Back-electromotive force, phase c	V	Real
o_omega_m	Rotor mechanical rotational speed	rad/s	Real
o_omega_e	Rotor electrical rotational speed	rad/s	Real
o_theta_m	Rotor mechanical position	rad	Real
o_theta_e	Rotor electrical position	rad	Real

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o_tau_e	Electromagnetic torque	Nm	Real
o_tau_shaft	Torque delivered to the shaft	Nm	Real
o_tau_rot_loss	Output Rotational Losses torque	Nm	Real
o_pe_b	Motor total power - calculated using power balance equation	W	Real
o_pe_dq	Motor total power - calculated by v*i in dq coordinates	W	Real
o_pe_abc	Motor total power - calculated by v*i in abc coordinates	W	Real
o_p_shaft	Power delivered to the shaft	W	Real
o_p_rot_loss	Output Rotational Losses	W	Real

Parameters			
Name	Description	Unit	Data type
Rs0	Initial stator winding resistance	Ohm	Real
Ld0	Initial stator inductance in the d axis	H	Real
Lq0	Initial stator inductance in the q axis	H	Real
phiPM1	Initial PM flux amplitude of the fundamental harmonic	Vs	Real
NPP	Number of pole pairs	-	Real
Jrotor	Rotor's inertia	kg*m2	Real
B	Friction coefficient	-	Real
Ts	Step size of the internal delay block (if used)	s	Real
TsRef	Reference temperature of the rotor (PM)	degC	Real
TrRef	Reference temperature of the stator	degC	Real
alpha_s	Stator resistance temperature coefficient	1/degC	Real
d_magnet	Magnets temperature coefficient	1/degC	Real
Rsa_aux	Additional stator winding resistance, phase a	Ohm	Real
Rsb_aux	Additional stator winding resistance, phase b	Ohm	Real
Rsc_aux	Additional stator winding resistance, phase c	Ohm	Real
Ls_Sat_enable	Stator inductance saturation enable	-	Real

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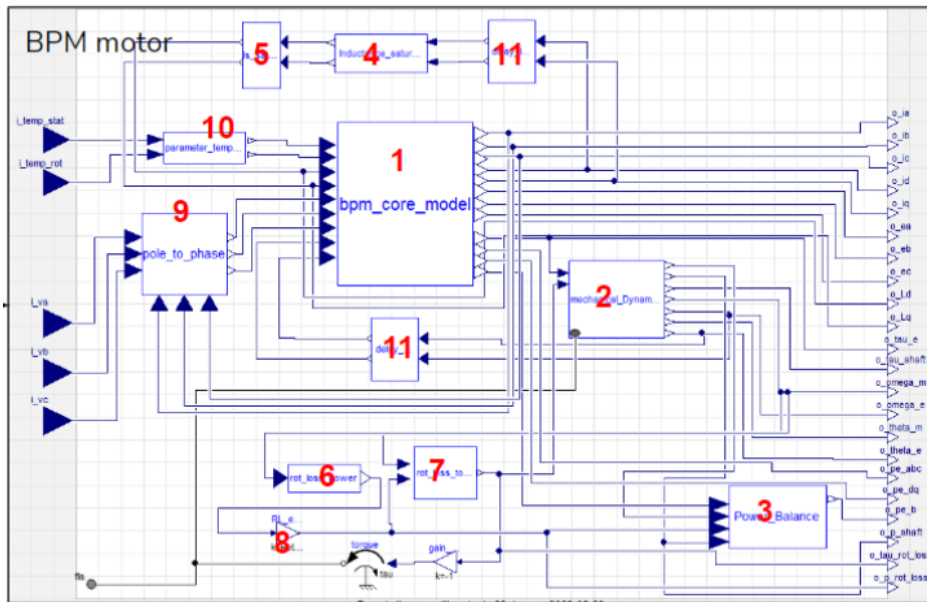
I_pk0 ... I_pk7	Stator inductance LUT input coefficients	A	Real
Ld_a0 ... Ld_a7	Stator inductance LUT coefficients to estimate $L_d = f(i_d)$	H	Real
Lq_b0 ... Lq_b7	Stator inductance LUT coefficients to estimate $L_q = f(i_q)$	H	Real
Rot_Loss_enable	Rotational losses calculation enable	-	Real
Rot_Loss_omega_th	Rotational losses calculation speed threshold	rad/s	Real
Rot_Loss_Speed0 ... Rot_Loss_Speed7	Rotational losses LUT input coefficients	rpm	Real
Rot_Loss_Power0 ... Rot_Loss_Power7	Rotational losses LUT coefficient to estimate $power_rot_loss = f(speed)$	W	Real

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4.3 MODEL FORMULATION

This document describes in detail only the complete BPM motor electro-mechanical model (1). Mechanical load models are not covered in this guideline - please refer to other available documentation. As shown on the diagram below, the Complete BPM model consists of several functional modules (components). Selected of them are described in detail in the following sections.



Functional modules:

- 1 - Core electrical model
- 2 - Mechanical dynamics
- 3 - Power balance
- 4 - Stator inductance variation
- 5 - Inductance variation enabler
- 6 - Rotational losses power
- 7 - Rotational losses torque
- 8 - Rotational losses enabler
- 9 - Pole to phase calculation
- 10 - Temperature variation
- 11 - Signal unit delay

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4.3.1 CORE ELECTRICAL MODEL (1)

A first-principles model describing the electrical dynamics of a motor. Essential outcomes are the motor currents and the generated electromagnetic torque. Also, this module calculates the total motor power and Joule losses. Motor equations are expressed in the DQ reference frame, therefore the Clarke-Park transformation of selected signals is required to apply, in order to transform the model reference frame from ABC to DQ coordinates.

$$\begin{bmatrix} f_d \\ f_q \\ f_0 \end{bmatrix} = 2/3 \cdot \begin{bmatrix} \cos(\theta_e) & \cos(\theta_e - \frac{2\pi}{3}) & \cos(\theta_e + \frac{2\pi}{3}) \\ -\sin(\theta_e) & -\sin(\theta_e - \frac{2\pi}{3}) & -\sin(\theta_e + \frac{2\pi}{3}) \\ 1/2 & 1/2 & 1/2 \end{bmatrix} \cdot \begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix}$$

The inverse transformation can be easily derived from the above equations. It is mainly required for calculating motor phase currents. As for now, the zero (f_0) component is not used in the model. Therefore, the implemented motor equations are based solely on the D and Q components:

- voltage equations

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} R_s & 0 \\ 0 & R_s \end{bmatrix} \cdot \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \psi_{sd} \\ \psi_{sq} \end{bmatrix} + \omega_e \begin{bmatrix} -\psi_{sd} \\ \psi_{sq} \end{bmatrix}$$

- permanent magnet flux equations (the PM flux is aligned to the D axis)

$$\begin{bmatrix} \psi_{md} \\ \psi_{mq} \end{bmatrix} = \begin{bmatrix} \psi_m \\ 0 \end{bmatrix}$$

- stator flux equations

$$\begin{bmatrix} \psi_{sd} \\ \psi_{sq} \end{bmatrix} = \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \cdot \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \psi_{md} \\ 0 \end{bmatrix}$$

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- currents dynamics

$$\frac{d}{dt} \begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} 1/L_d(v_d - R_s i_d + \omega_e L_q i_q) \\ 1/L_q(v_q - R_s i_q - \omega_e L_d i_d \psi_m) \end{bmatrix}$$

* R_s , ψ_m - can vary due to temperature,
 * L_d , L_q - can vary due to the saturation effect.

- electromagnetic torque generation

$$\tau_e = 3/2 \cdot NPP \cdot (\psi_{sd} i_q - \psi_{sq} i_d)$$

- power calculations
 - total power in DQ reference frame
 - total power in ABC reference frame (more as a debug signal)

$$p_{edq} = 3/2 \cdot (v_d i_d + v_q i_q)$$

$$p_{eabc} = (v_a i_a + v_b i_b + v_c i_c)$$

- Joule power losses

$$p_j = i_a^2 R_s + i_b^2 R_s + i_c^2 R_s$$

4.3.2 MECHANICAL DYNAMICS (2)

A first-principles model to calculate:

- mechanical rotor speed and position

$$\frac{d\omega_m}{dt} = \frac{1}{J_r} (\tau_e - \tau_l - B\omega_m)$$

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$$\frac{d\theta_m}{dt} = \omega_m$$

- electrical rotor speed and position

$$\omega_e = NPP \cdot \omega_m$$

$$\theta_e = NPP \cdot \theta_m$$

- mechanical torque and power delivered to the shaft

$$\tau_{sh} = \tau_e - \tau_{RL}$$

$$p_{sh} = \tau_{sh} \cdot \omega_m$$

4.3.3 POWER BALANCE (3)

This module calculates the total power of the motor based on the power balance equation. According to theoretical considerations, the output power from this block should be equal to the total power calculated by the core model (1).

$$p_{eb} = p_j + p_{sh} + p_{RL}$$

4.3.4 STATOR INDUCTANCE VARIATION (4)

Stator inductance may vary during motor operation e.g. due to a saturation effect. An empirical approach is used to model this phenomenon. The approach relies on look up tables (LUTs) which calculate each inductance based on coefficients $I_{pk0} \dots I_{pk7}$, $L_{d_a0} \dots L_{d_a7}$, $L_{q_b0} \dots L_{q_b7}$ which are derived by identification procedures performed on the laboratory test bench. Inductance values in D and Q axis are calculated separately based on instantaneous i_d and i_q currents values respectively.

In general, inductances can be described as $L_d = f(|i_d|)$, $L_q = f(|i_q|)$ where the function f is a linear interpolation of LUT points.

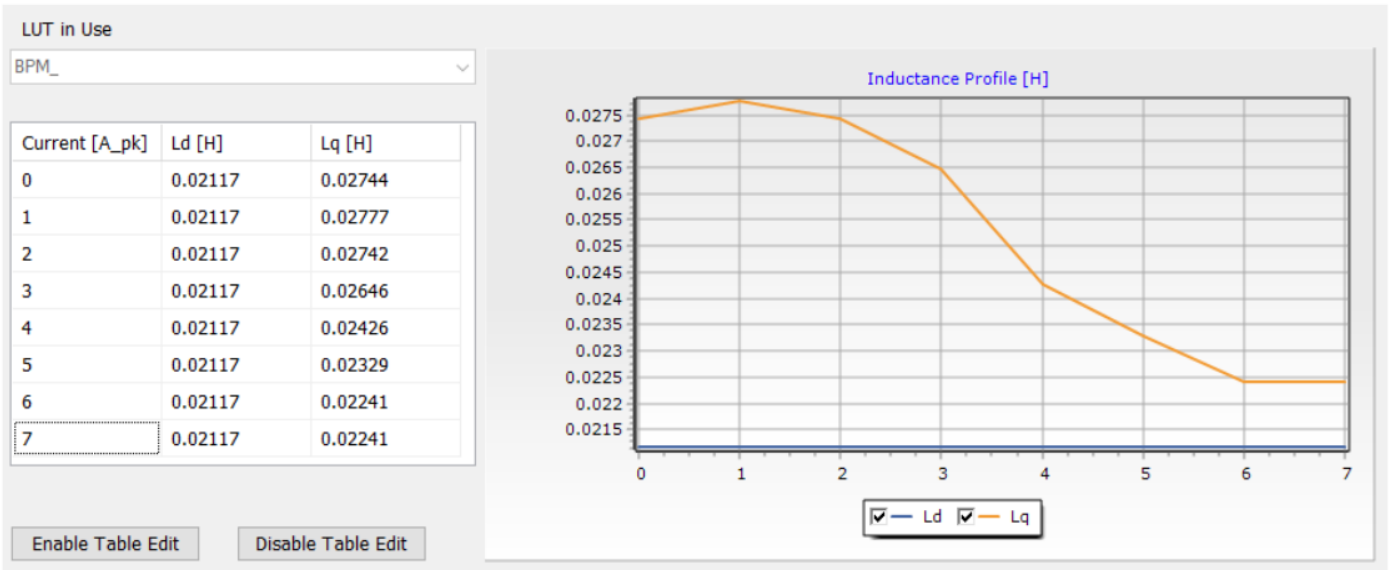
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I_{pk} [A]	L_d [H]	I_{pk} [A]	L_q [H]
I_{pk0}	L_{d_a1}	I_{pk0}	L_{q_b1}
I_{pk1}	L_{d_a2}	I_{pk1}	L_{q_b2}
I_{pk2}	L_{d_a3}	I_{pk2}	L_{q_b3}
I_{pk3}	L_{d_a4}	I_{pk3}	L_{q_b4}
I_{pk4}	L_{d_a5}	I_{pk4}	L_{q_b5}
I_{pk5}	L_{d_a6}	I_{pk5}	L_{q_b6}
I_{pk6}	L_{d_a7}	I_{pk6}	L_{q_b7}
I_{pk7}	L_{d_a8}	I_{pk7}	L_{q_b8}

- L_d, L_q calculation example

Inductance LUT



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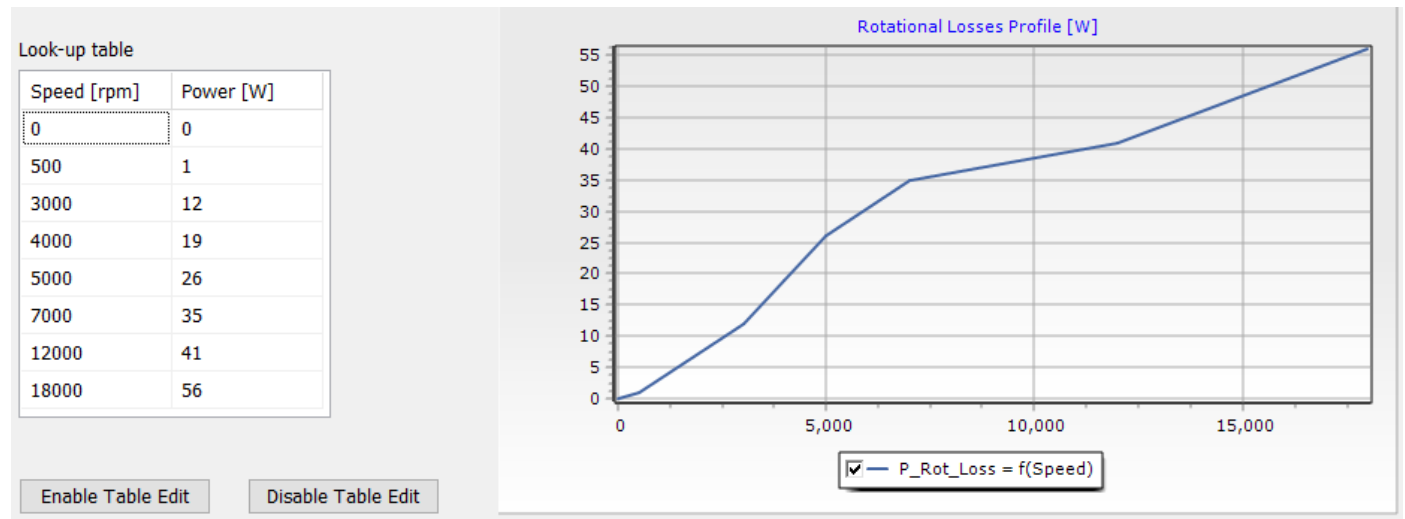
4.3.5 ROTATIONAL LOSSES POWER (6)

Rotational losses power calculation also relies on the lookup table approach. In this case, the coefficients are also derived from test procedures performed on the laboratory test bench.

In general, the rotational losses power can be described as $p_{RL} = f(|n|)$ where the function is the linear interpolation of rotational losses LUT coefficients.

n [rpm]	p_{RL} [W]
n_0	p_{RL0}
n_1	p_{RL1}
n_2	p_{RL2}
n_3	p_{RL3}
n_4	p_{RL4}
n_5	p_{RL5}
n_6	p_{RL6}
n_7	p_{RL7}

- rotational losses power calculation example

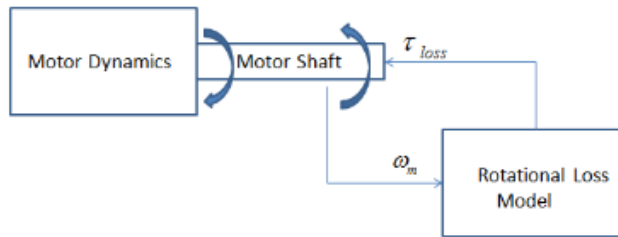


4.3.6 ROTATIONAL LOSSES TORQUE (7)

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Loss torque is considered as opposing torque in the mechanical dynamics of the motor as shown on the figure.



It is derived from:

$$\tau_{RL} = \frac{p_{RL}}{\omega_m}$$

4.3.7 POLE TO PHASE VOLTAGE CALCULATION (9)

This module reconstructs the pole into the phase voltage passed to the core electrical BPM model. Also, it enables to simulate phase unbalance conditions by injecting additional resistances to each phase separately. Picture below shows a typical diagram of connecting a BPM motor with a voltage source inverter.

From a standpoint of the BPM input voltage, this electrical circuit can be described following equations:

$$V_n = \frac{1}{3} (V_a + V_b + V_c - i_a \cdot R_{saux} - i_b \cdot R_{sbux} - i_c \cdot R_{scaux})$$

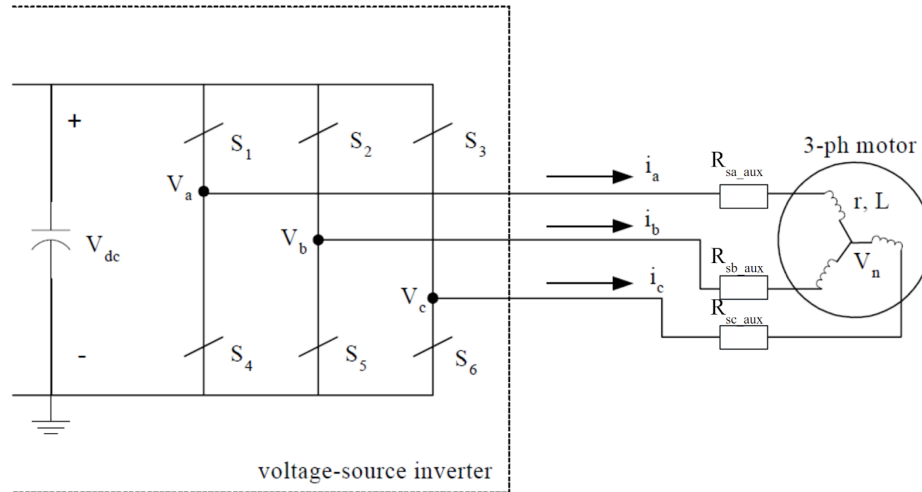
$$V_{an} = V_a - V_n$$

$$V_{bn} = V_b - V_n$$

$$V_{cn} = V_c - V_n$$

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where:

- V_a, V_b, V_c - pole voltages
- V_{an}, V_{bn}, V_{cn} - phase voltages
- V_n - potential in the neutral point
- i_a, i_b, i_c - motor phase currents

4.3.8 TEMPERATURE VARIATION (10)

This module calculates changes of the stator resistance and permanent magnet flux amplitude due to stator and rotor temperature variations. The outputs are multiplication coefficients (k_{temp_Rs} , k_{temp_phiPM}) passed to the core electrical model. Input temperatures are converted from degC to K.

$$k_{tempRs} = (1 + \alpha_s) * ((Ts + 273.15) - (Ts_{Ref} + 273.15))$$

$$k_{tempPM} = (1 - \alpha_M) * ((Tr + 273.15) - (Tr_{Ref} + 273.15))$$

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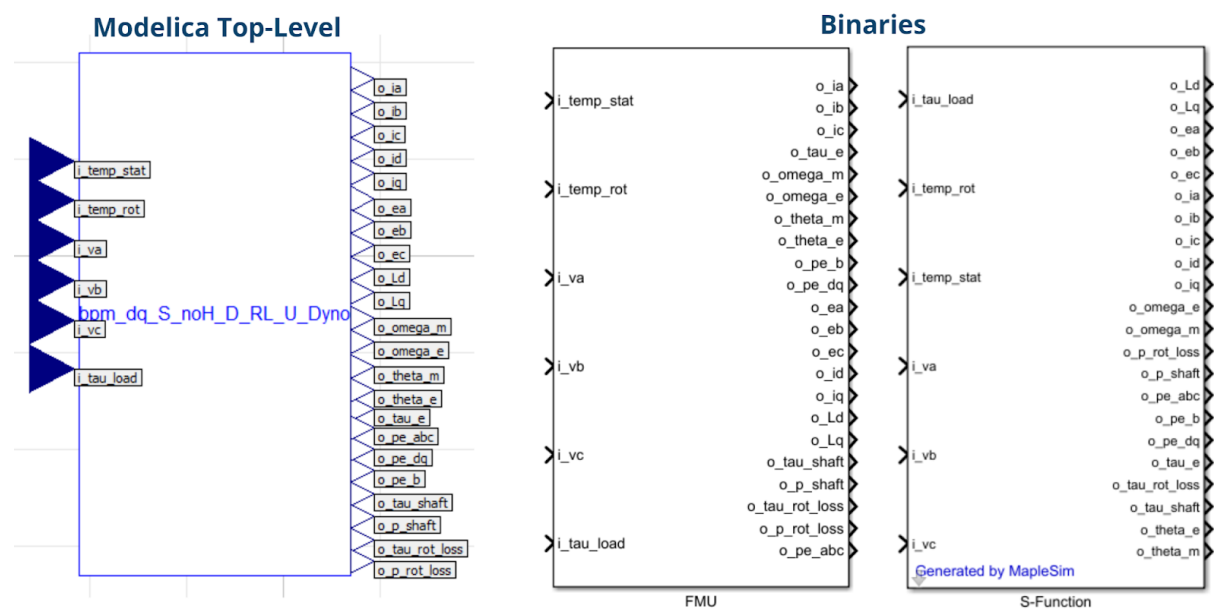
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5. INTEGRATION OUTSIDE MODELICA ENVIRONMENT

In order to integrate the model outside the Modelica environment, a binary needs to be generated. Two basic binary forms are used in the case of the BPM model: FMU and S-Function.

5.1 CORE INTERFACE

After binary generation the core interface remains the same as in the top-level Modelica model (section 6.2). All model inputs, outputs and parameters are exposed.



5.2 INTEGRATION IN THE ELECTRIC DRIVE LIBRARY

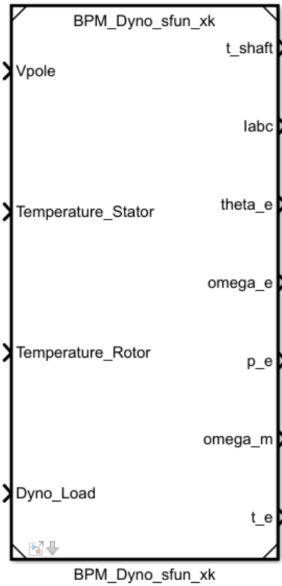
Ultimately the model is embedded into the Electric Drive Library. Due to specific requirements, the model is integrated there by using customized methods.

In the Matlab/Simulink environment, the Simulink Top-Level models have an optimized interface, limited only to essential signals. Binaries are integrated at the Low-Level. In terms of models parameterization, customized Top-Level block masks have been designed as well.

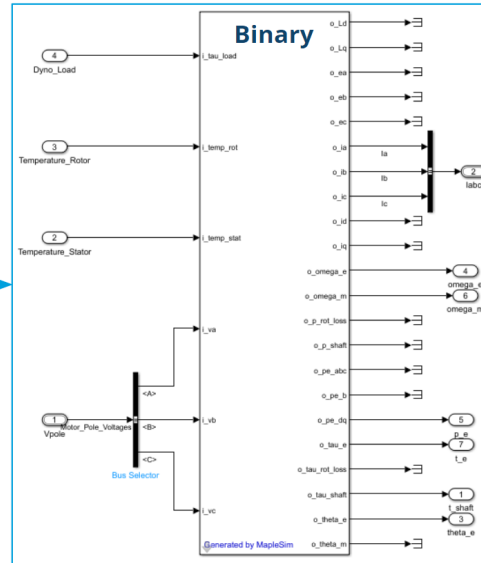
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Simulink Top-Level



Simulink Low-Level



Top-Level Model Mask

Block Parameters: BPM_Dyno_sfunk_xk

BPM_Dyno_sfunk_xk (mask)

Three phase Brushless Permanent Magnet motor.

Transient electro-mechanical motor model coupled with dyno.

Discrete-time model with adjustable time step*.

*Model time step has to be predefined before simulation in a workspace variable TimeSteps.ModelStep.

Basic motor parameters

Electrical Mechanical Thermal

Initial stator winding resistance [Ohm] Motor_prm.Electrical.Rs_0

Initial inductance in the D axis [H] Motor_prm.Electrical.Ld_0

Initial inductance in the Q axis [H] Motor_prm.Electrical.Lq_0

Initial magnet flux (1st harmonic) [V*s] Motor_prm.Electrical.PM_flux_h1_0

Number of pole pairs [-] Motor_prm.Electrical.NPP

Advanced motor parameters

Inductance saturation Power losses Current unbalance

Inductance Saturation flag [0/1]

1

*LUT coefficients have to be predefined before simulation in a workspace structure Motor_prm.Inductance_Variation

*Input current [A]

Motor_prm.Inductance_Variation.Ipk

*Inductance LUT coef. to calculate $L_d = (I_{pk})$ [H]

Motor_prm.Inductance_Variation.Ld_a

*Inductance LUT coef. to calculate $L_q = (I_{pk})$ [H]

Motor_prm.Inductance_Variation.Lq_b

OK Cancel Help Apply

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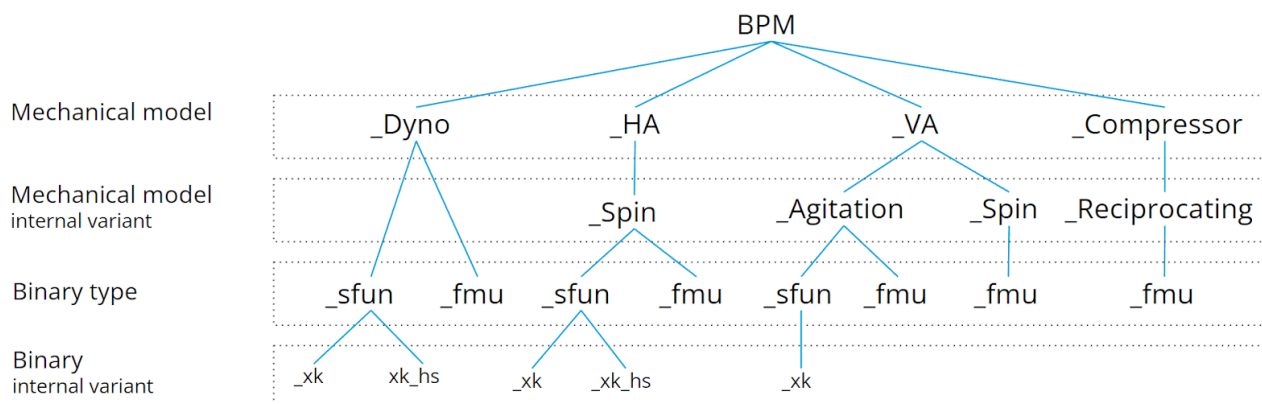
Source Document: W10335242 Rev E

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	Description (Local)		Revision B
Whirlpool Corporation, Benton Harbor, MI, USA			

6. MODEL VARIANTS

The BPM motor model has several core, top-level variants, which are mainly determined by:

- coupled mechanical load (Dyno, VA, HA, Compressor),
- binary form (FMU, S-Function),
- special use case (e.g. high speed range, PM flux harmonics).



CONTRIBUTORS

(Individuals who contributed to the content of this document)

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REVISION HISTORY

REV	DESCRIPTION OF CHANGE	SECTION	BY	CN	DATE
A	Draft	All	KIRTACA	-	01-Dec-2017
B	Major update. Document reorganization. Alignment with the latest BPM transient model (v2.5.0).	All	BEDNASA	-	06-Feb-2023

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